ECSE 331
Laboratory No. 2
Simulation of semiconductor diode in PSPICE

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***Abstract*—A diode is a two-terminal semiconductor device that conducts current primarily in one direction. Despite being simple two-pin devices, their applications such as voltage rectification, signal mixing, signals isolation, and lighting make them vital in electronics. Diodes have become an essential part of all electronics. As the complexity of electronic devices increases, designing circuits by hand becomes an impossible task due to being very time-consuming and prone to human errors. Consequently, engineers design intricate circuits using circuit simulation software such as SPICE. In this laboratory, we design and simulate diode circuits to analyze their characteristics and some of their applications in electronic circuits. This includes analyzing diode specifications such as the cut-off voltage, Zener breakdown, conduction in forward bias, and the effect of temperature on diode characteristics. We then build and simulate circuits using diodes, namely a full-wave bridge rectifier and a clipper circuit.**

I. Introduction

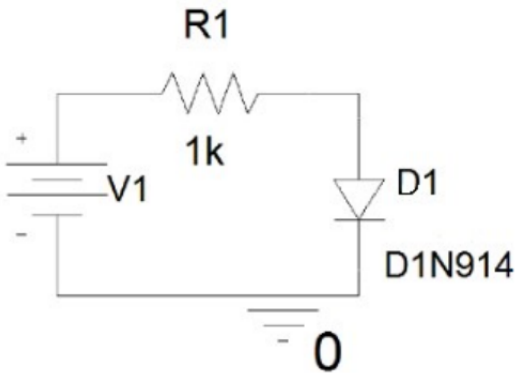
A diode is a two-terminal semiconductor device that conducts current primarily in one direction. It is essentially a one-way current switch. It has a low resistance in one direction and high resistance in the other. The diode allows current to flow in forward-biased whereas, in reverse-bias, the diode blocks the current flow.

In this laboratory, we conduct experiments on five circuits to get familiarized with the use of simulation software in designing and analyzing circuits consisting of diodes. We first study the cut-off voltage of a silicon diode and investigate the effect that temperature has on it. We then build and analyze three different circuits that show the application of diodes, particularly two different full-wave bridge rectifiers and a clipper circuit. Finally, we study some of the characteristics of a Zener diode

Task 1

This experiment required using PSPICE to build the circuit shown in Figure 1 and to find the cut-off voltage of the diode by plotting the variation of current through the diode with respect to the voltage across it.

Fig. 1. Task 1 circuit



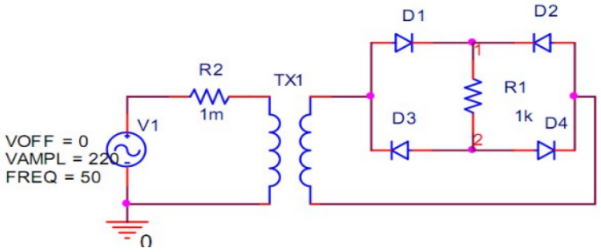
Task 2

In this experiment, we study the effect of temperate on diode characteristics by finding the I-V plot at 50 °C, 100 °C, and 150 °C.

Task 3

In this experiment, we construct a full-wave bridge rectifier circuit shown in Figure 2. We then have to plot the voltage V1 and the output voltage across R1 versus time.

Fig. 2. Full Wave bridge rectifier



Task 4

In this experiment, we add a capacitor in parallel to the R1 resistor in the circuit shown in Figure 2. Again, we plot the output voltage across R1 across time by experimenting with three different capacitor values.

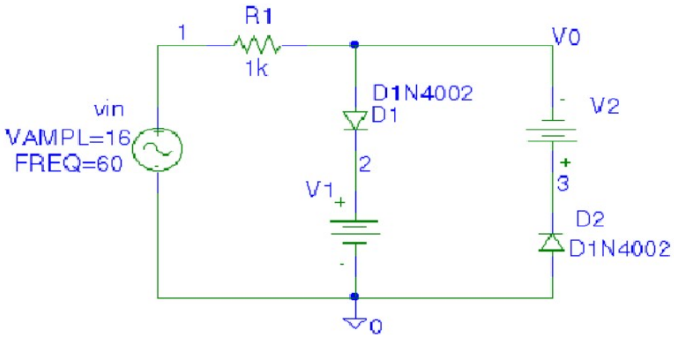
Task 5

In this experiment, we build the clipper circuit shown in Figure 3. We choose the values of V1 and V2 such that the output voltage V0 is as follows:

V0 = Vin for -5V<Vin<9V
V0 = 9V for Vin >9V
V0 = -5V for Vin<-5V

This simulation is then followed by a theoretical discussion.

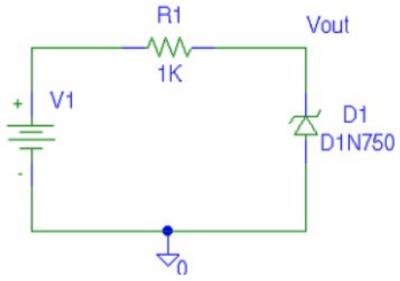
Fig. 3. Clipper circuit



Task 6

In this experiment, we construct the circuit shown in Figure 4. We plot the variation of current across the Zener diode with respect to the variation of voltage to find the voltage at which the Zener breakdown occurs and the voltage at which the diode starts to conduct in the forward bias region.

Fig. 4. Zener diode circuit

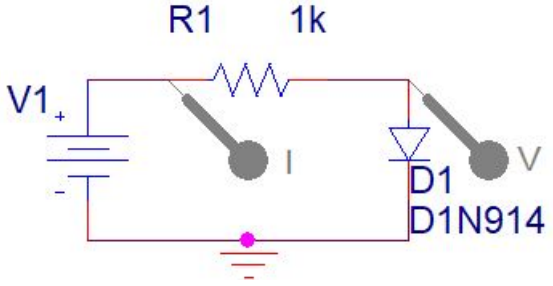


II. Analysis

Task 1

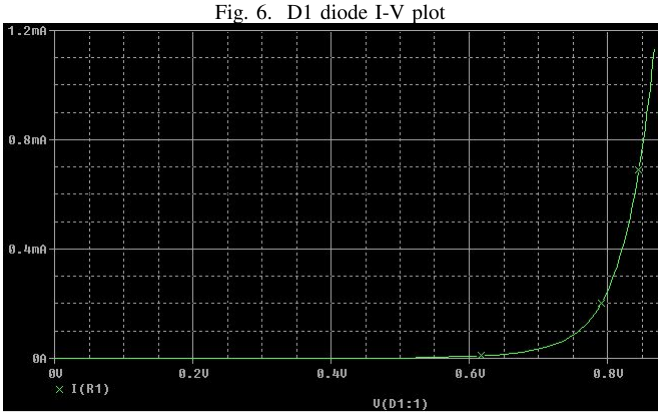
We first construct the circuit showed in Figure 1 using an independent DC voltage source, a resistor, and a D1N914 diode. Our PSIPCE schematic for this circuit is shown in Figure 5. Once the schematic is made, we perform a DC sweep simulation on the circuit. We sweep the value of the voltage source V1 from -2V to 2V. We place a current and a voltage probe in positions shown in Figure 5, allowing us to plot the I-V curve of the diode. We then plot the current across the diode D1 and set the x-axis of the plot as the voltage across the diode.

Fig. 5. Task 1 circuit schematic



The variation of current through the diode with respect to the voltage across it can be seen from the I-V plot shown in Figure 6.

Fig. 6. D1 diode I-V plot



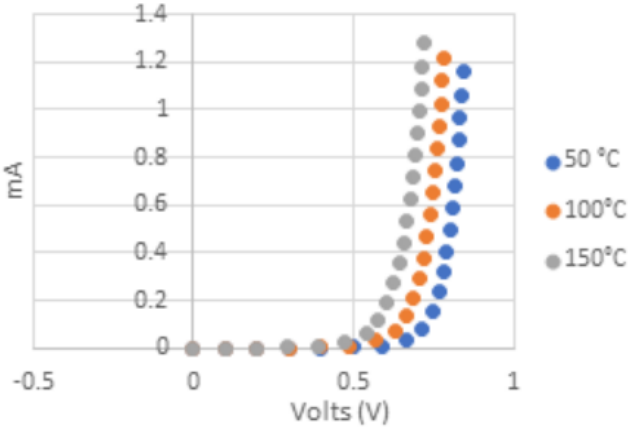
The cut-in voltage is found from the voltage at which the diode starts to conduct (has a positive current). This voltage is at 0.65V. We find the cut-off voltage to be approximately 0.8V when the diode, is fully conducting but remains at an approximately constant voltage. Therefore, $V_{cut-in} \approx 0.65V$ and $V_{cut-off} \approx 0.8V$.

Task 2

Using the circuit schematic previously build and explained in Figure 5, we run a simulation to obtain the I-V plot of the diode at three different temperatures: 50 °C, 100 °C, and 150 °C. We perform this simulation by simply adding a temperature sweep for each of the mentioned temperatures to our simulation profile.

The superposed I-V plot (current through the diode with respect to the voltage across the diode) for each of the three temperatures can be seen in Figure 7.

Fig. 7. D1 diode I-V plot at 50 °C, 100 °C, and 150 °C



We can see that the I-V plot of the diode changes with temperature from Figure 7. We observe that both the cut-off and cut-in voltages decrease as temperature rises. PN junction diodes parameters such as the cut-in and cut-off voltage, the reverse breakdown, and barrier voltage are all highly dependant on the temperature.

The diode current is given using the Shockley equation given bellow.

$$I = I_S(e^{\frac{V_D}{nV_T}-1}) \tag{1}$$

$$V_T = \frac{kT}{q} \tag{2}$$

where I is the diode current,
I_S is the reverse saturation current,
V_D is the voltage across the diode,
V_T is the thermal voltage,
n is the ideal factor

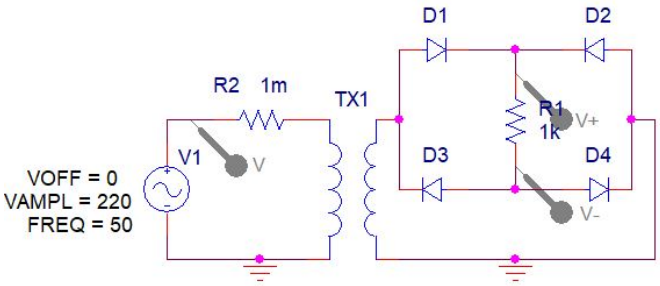
Therefore, according to the Shockley diode equation, we conclude that the current should decrease with an increase in temperature. However, according to our observations from Figure 7, the exact opposite happens. This can be explained by the fact that with the rise of temperature, more electron-hole pairs are created resulting in higher conductivity and

reverse saturation current. The reverse saturation current I_S of a silicon diode increase by 7%/°C. Therefore, a decrease in the cut-in and cut-off voltage with the rise of temperature. Reverse breakdown voltage (VR) also increases as we increase the temperature.

Task 3

A full-wave rectifier circuit produces an output voltage or current which is either completely DC or has some particularized DC component. In this exercise, we build and simulate a full-wave bridge rectifier. In our design of this single-phase rectifier, we use four D1N914 rectifying diodes connected in a closed-loop bridge. Additionally, the circuit is comprised of two resistors, an AC source with an amplitude of 220V and a frequency of 50Hz, and a transformer with the coupling set to 1, $L1_{value} = 100\mu H$ and $L2_{value} = 1\mu H$. The schematic of our design can be seen in Figure 8.

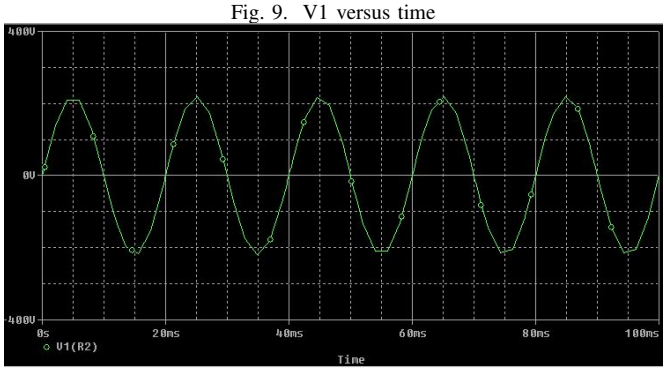
Fig. 8. Full wave bridge rectifier circuit schematic



Our main goal is to obtain the input and output voltages of the full-wave bridge rectifier. For this reason, we place a voltage probe at the AC source and two voltage probes to find the voltage drop across the resistor R1. We perform a time-domain (transient) simulation of 100 ms.

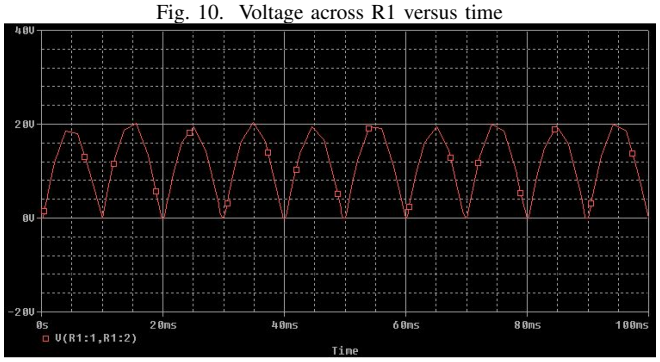
As mentioned before, the V1 has an amplitude of 220V with a frequency of 50Hz. This can be confirmed from Figure 9 showing the obtained plot for the input voltage V1 versus time.

Fig. 9. V1 versus time



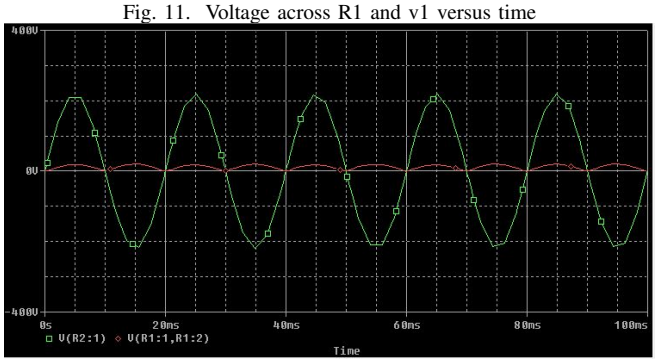
The plot of the output voltage (the voltage across the resistor R1) versus time was is obtained and shown in Figure 10. From Figure 10, we can see a full-wave rectification where the voltage oscillates between 0 and 20V. By simple circuit analysis, this rectification can be explained by the fact that during the positive half cycle of the supply, diodes D1 and D4 conduct in series conduct is series but diodes D2 and D3 are off as they are reverse biased. However, during the negative half cycle, diodes D2 and D3 conduct in series while diodes D1 and D4 switch off due to being in reverse bias. So the current flow through R1 will be in the same directions.

Fig. 10. Voltage across R1 versus time



To better visualize the full wave rectification, Figure 11 shows V1 and the voltage across R1 versus time in a superposed manner.

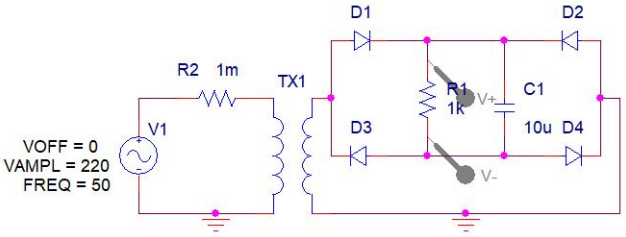
Fig. 11. Voltage across R1 and v1 versus time



Task 4

As we had an AC input source (V1), the output of the full-wave bridge rectifier is a polarized, pulsating, and non-sinusoidal signal. It has $1/10^{th}$ the amplitude of the input signal with double the frequency of the input.

Fig. 12. Full wave bridge rectifier with smoothing



The output signal can be viewed as a DC voltage upon which a very large ripple voltage is superimposed. This output,

found in task 4 is not very efficient as the ripple is dissipated as heat in DC components and causes noise in operations. To get a more suitable output signal, we have to convert all the ripple voltage, seen in Figure 10, into smoother and higher DC output. In this exercise, we add a capacitor in parallel to the resistor R1 in the circuit in Figure 8. The new circuit schematic is presented in Figure 12.

We simulate the schematic shown in Figure 12, under the same settings and conditions as Task 3. We first set the value of the capacitor C1 to be $10\mu F$, we then increase the capacitance by setting it to $100\mu F$ and $1mF$, each time, plotting the voltage across R1 versus time. The resulting plots can be seen in Figures 13, 14, and 15.

Fig. 13. Voltage across R1 vs time with $C = 10\mu F$

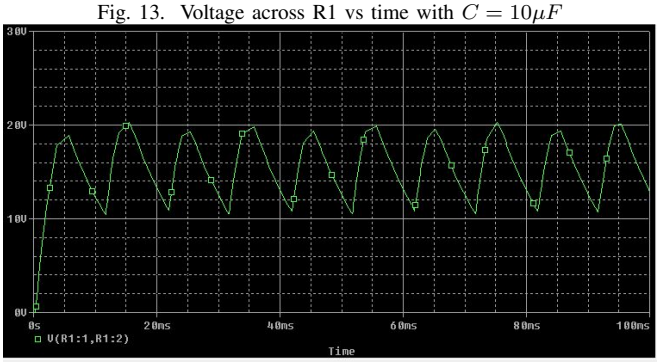


Fig. 14. Voltage across R1 vs time with $C = 100\mu F$

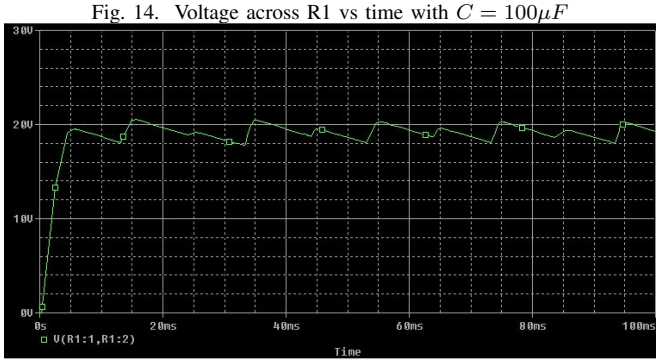
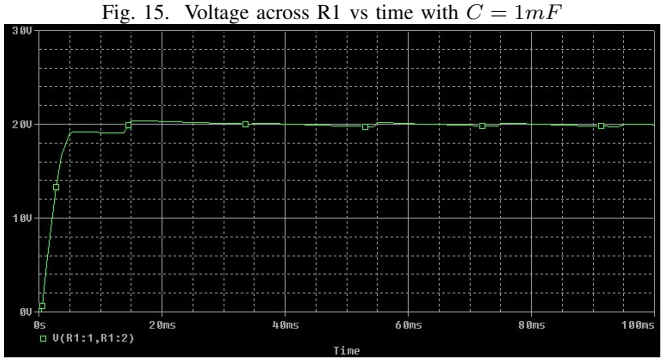


Fig. 15. Voltage across R1 vs time with $C = 1mF$



We can conclude from Figures 13, 14, and 15 that increasing the capacitance of the capacitor C1 reduces the ripple. With $C = 1mF$, we have converted the ripple voltage into a smoother signal of 20V DC.

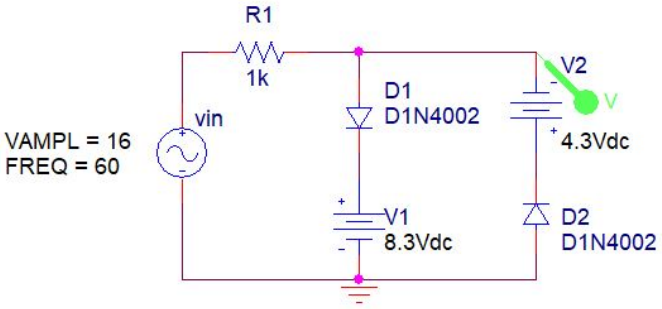
This smoothing can be explained by the fact that with a large enough capacitor ($1mF$ in our case) when the ripple voltage rises, reactive power is stored in the capacitor. When the ripple voltage falls, reactive power is discharged from the capacitor raising the voltage. This also explains the reason why smaller capacitors did not fully remove the ripple voltage.

Task 5

A clipper is a circuit designed to block a signal from exceeding a certain reference voltage level without distorting the remainings of the signal. In this exercise, we build a dual diode clipper that combines a parallel negative clipper with a negative bias with a positive clipper with a positive bias. This configuration, shown in Figure 3, allows us to clip at both ends of the signal.

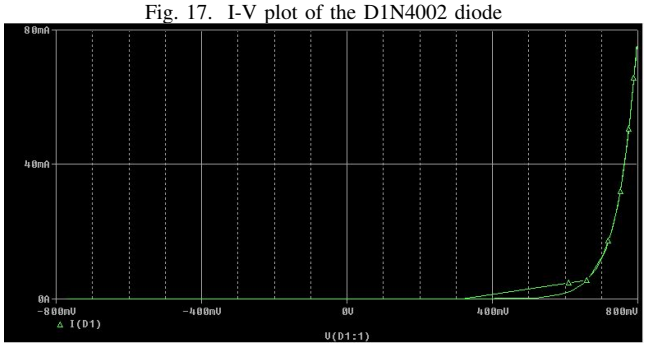
We build the Figure 3 circuit using an AC power source with an amplitude of 16 and a frequency of 60Hz, one resistor, two DC voltage sources, and two D1N4002 diodes as shown in Figure 16.

Fig. 16. Clipper circuit schematic



To find the voltages for the two independent voltage sources V1 and V2, we first need to find the cut-off voltage of the D1N4002 diode similar to what we did in task 1 for the D1N914 diode. So we plot the current going through the diode against the voltage across the diode. In our simulation, we use a voltage step of 0.01V. The I-V plot found for the D1N4002 diode is shown in Figure 17. We can infer the cut-off voltage to be approximately equal to 0.67V from Figure 17 as the current increases significantly past that voltage.

Fig. 17. I-V plot of the D1N4002 diode



Knowing the cut-off voltage of the diode, we can conduct our analysis to find the voltages V1 and V2 of the two independent DC sources such that the clipper meets the design requirements mentioned previously. We can solve this circuit by conducting the constant voltage drop model. The two diodes can both be off or on or one diode can be on while the other is off. Four cases are as follows:

1) When both D1 and D2 are on, we get contradictory results
2) D1 and D2 are both off

$$\Rightarrow V_{in} = Vout$$

3) D2 is off and D1 is on

$$\Rightarrow V1 = V_{max-out} - V_{D1} = 9 - 0.7 = 8.3V$$

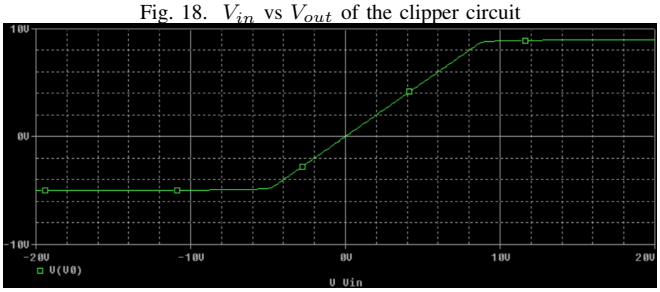
4) D1 is off and D2 is on

$$\Rightarrow V2 = V_{min-out} + V_{D2} = -5 + 0.7 = -4.3V$$

For simplification, we rounded V_D up to 0.7V. From the above equations, we find that V1=8.3V and V2=-4.3V.

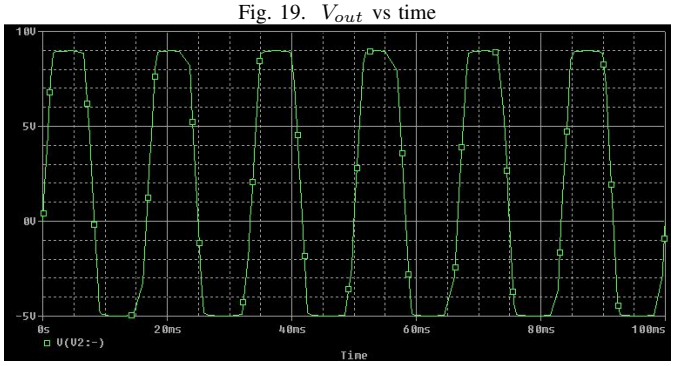
To verify our choices of voltage sources V1 and V2, we perform a simulation to plot Vin vs $Vout$. The result of this simulation is shown in Figure 18. We can see that our design specifications have been correctly met.

Fig. 18. V_{in} vs V_{out} of the clipper circuit



Additionally, we can check the output voltage versus time by carrying a time-domain simulation over 100ms. The result of this simulation is shown in Figure 19. We can see that with values found for V1 and V2, the voltage stays between -5V and 9V. Moreover, we can see that the voltage gets clipped outside the [-5V, 9V] bound.

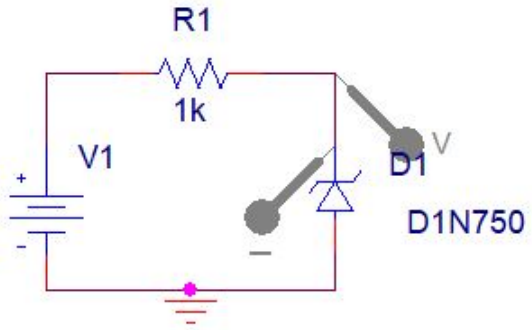
Fig. 19. V_{out} vs time



Task 6

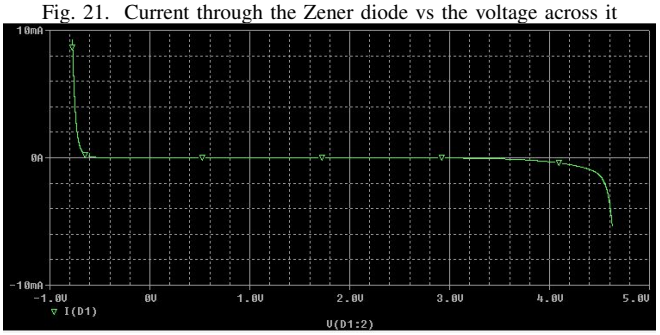
In this exercise, we design the circuit given in Figure 4 using the D1N750 Zener diode. The circuit schematic created in SPICE can be seen in Figure 20.

Fig. 20. Task 6 circuit schematic



We place a voltage and current probe to find the current through the diode and the voltage across it. We then perform a sweep simulation to plot the current through the Zener diode against the voltage across it. The resulting plot, from this simulation, is shown in Figure 21.

Fig. 21. Current through the Zener diode vs the voltage across it



From this Figure 21 plot, we can find that the voltage at which Zener breakdown occurs is about 4.5V and the voltage at which the diode starts to conduct in forward bias is close to -0.65V.

III. Conclusion

To conclude, in this laboratory, we performed a series of simulations and analyses on diodes and used them to construct circuits with noteworthy applications. We analyzed the characteristics of the D1N914 diode and the D1N750 Zener diode. We performed studies on the effect of temperature on diode characteristics. We then focused on leveraging studied diode characteristics to construct a full wave bridge rectifier, to which we then added a capacitor to achieve smoothing. Finally, we designed, simulated, and analyzed a dual diode clipper circuit.